Comparative Research on GPU, Its Operating Systems, and Their Application in AWS Cloud Platforms

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1. Introduction

Graphics Processing Units (GPUs) have been developed, which are no longer straightforward graphics accelerators but have significantly expanded in terms of their parallelization power to thousands of threads being executed at the same time. The extensive application of GPUs in scientific computing, machine learning, and artificial intelligence, as well as cloud computing systems, has highlighted the importance of comprehending how to operate them or how they behave at both the hardware and software levels. The modern GPUs work under extremely specialized drivers and interfaces, which are services of the underlying operating system that allow the developer to communicate with the hardware of GPUs in order to both render graphics and do some general-purpose computing.

This coursework primarily aims to perform a comparative research study on GPUs and their interrelationship with Linux operating systems and command-line utilities and then to implement a cloud-based solution based on the AWS platform. In this work, the elements of theoretical examination and practical experimentation indicate the use of GPU monitoring, diagnostics, and management of Linux systems and the ways in which these systems can be implemented or operated within cloud systems.

In this report, I have presented the following four coursework tasks as follows:

- Task 1: Learn to use Linux GPU operating system command-line tools in monitoring and diagnostics.
- Task 2: Hands-on work on the EC2 instances on the Amazon cloud.
- Task 3: The analysis of the tools of the command line that can be used as applied to cloud computing.
- Task 4: Conclusion of summative findings and results.

The main goals of Task 1 included identifying and using five Linux GPU command-line tools to trace the work of the graphics card, load kernel modules, and identify a device in a virtualized environment. Task 2 was about the launching, configuring, and managing of AWS EC2 instances, which has provided an opportunity to have a hands-on experience in cloud deployment, monitoring, and managing of resources. The implementation, results, and discussion of each of the tasks are given in the following sections.

2. Task 1: Operating Systems – Linux GPU Command-Line Tools

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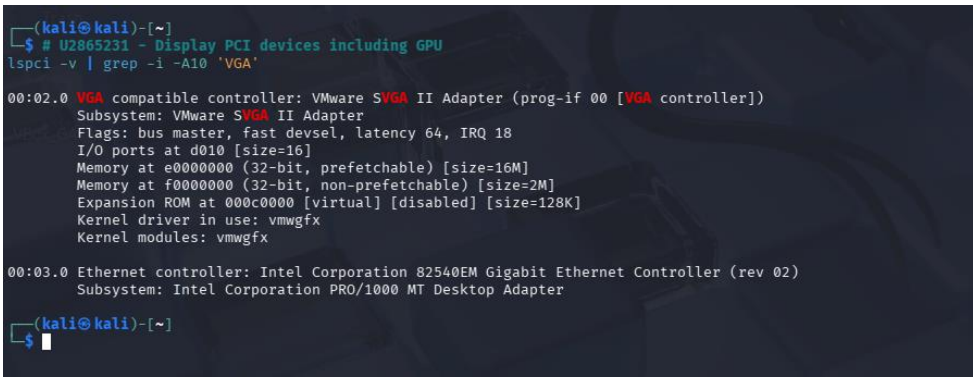

```
┌──(kali㉿kali)-[~]
└─$ # U2865231 - Display PCI devices including GPU
lspci -v | grep -i -A10 'VGA'

00:02.0 VGA compatible controller: VMware SVGA II Adapter (prog-if 00 [VGA controller])
        Subsystem: VMware SVGA II Adapter
        Flags: bus master, fast devsel, latency 64, IRQ 18
        I/O ports at d010 [size=16]
        Memory at e0000000 (32-bit, prefetchable) [size=16M]
        Memory at f0000000 (32-bit, non-prefetchable) [size=2M]
        Expansion ROM at 000c0000 [virtual] [disabled] [size=128K]
        Kernel driver in use: vmwgfx
        Kernel modules: vmwgfx

00:03.0 Ethernet controller: Intel Corporation 82540EM Gigabit Ethernet Controller (rev 02)
        Subsystem: Intel Corporation PRO/1000 MT Desktop Adapter

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┌──(kali㉿kali)-[~]
└─$ # U2865231 - Show loaded GPU-related kernel modules
lsmod | grep vmwgfx

vmwgfx                491520  2
drm_ttm_helper         16384  2 vmwgfx
ttm                   135168  2 vmwgfx,drm_ttm_helper
drm_client_lib         12288  1 vmwgfx
drm_kms_helper        258048  3 vmwgfx,drm_ttm_helper,drm_client_lib
drm                   831488  8 vmwgfx,drm_kms_helper,drm_ttm_helper,drm_client_lib,ttm

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┌──(kali㉿kali)-[~]
└─$ # U2865231 - Display current DRM/GPU devices detected by the kernel
ls /sys/class/drm/

card0           card0-Virtual-2  card0-Virtual-4  card0-Virtual-6  card0-Virtual-8  version
card0-Virtual-1 card0-Virtual-3  card0-Virtual-5  card0-Virtual-7  renderD128

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┌──(kali㉿kali)-[~]
└─$ # U2865231 - Display GPU-related device nodes
ls -l /dev/dri/

total 0
drwxr-xr-x  2 root root         80 Jan  3 15:43 by-path
crw-rw----+ 1 root video  226,   0 Jan  3 15:48 card0
crw-rw----+ 1 root render 226, 128 Jan  3 15:48 renderD128

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┌──(kali㉿kali)-[~]
└─$ # U2865231 - Show kernel messages related to GPU/graphics driver
dmesg | grep -iE 'drm|vmwgfx|gpu'

[    2.431457] ACPI: bus type drm_connector registered
[    2.843465] vmwgfx 0000:00:02.0: vgaarb: deactivate vga console
[    2.848678] vmwgfx 0000:00:02.0: [drm] FIFO at 0x00000000f0000000 size is 2048 KiB
[    2.848693] vmwgfx 0000:00:02.0: [drm] VRAM at 0x00000000e0000000 size is 16384 KiB
[    2.848703] vmwgfx 0000:00:02.0: [drm] Running on SVGA version 2.
[    2.848710] vmwgfx 0000:00:02.0: [drm] Capabilities: rect copy, cursor, cursor bypass, cursor bypass 2, alpha cursor, extended fifo, pitchlock, irq mask
, gmr, traces, gmr2, screen object 2, command buffers, command buffers 2, gbobject,
[    2.848713] vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported hypervisor.
[    2.848720] vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
[    2.848724] vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device to avoid problems.
[    2.848728] vmwgfx 0000:00:02.0: [drm] DMA map mode: Caching DMA mappings.
[    2.848836] vmwgfx 0000:00:02.0: [drm] Legacy memory limits: VRAM = 16384 KiB, FIFO = 2048 KiB, surface = 507904 KiB
[    2.848840] vmwgfx 0000:00:02.0: [drm] MOB limits: max mob size = 131072 KiB, max mob pages = 4096
[    2.848842] vmwgfx 0000:00:02.0: [drm] Max GMR ids is 8192
[    2.848844] vmwgfx 0000:00:02.0: [drm] Max number of GMR pages is 1048576
[    2.848845] vmwgfx 0000:00:02.0: [drm] Maximum display memory size is 16384 KiB
[    2.852076] vmwgfx 0000:00:02.0: [drm] Screen Target display unit initialized
[    2.852379] vmwgfx 0000:00:02.0: [drm] Fifo max 0x00200000 min 0x00001000 cap 0x00000355
[    2.853055] vmwgfx 0000:00:02.0: [drm] Using command buffers with DMA pool.
[    2.853075] vmwgfx 0000:00:02.0: [drm] Available shader model: Legacy.
[    2.854538] [drm] Initialized vmwgfx 2.21.0 for 0000:00:02.0 on minor 0
[    2.857699] fbcon: vmwgfxdrmfb (fb0) is primary device
[    2.862818] vmwgfx 0000:00:02.0: [drm] fb0: vmwgfxdrmfb frame buffer device
[    5.305706] systemd[1]: modprobe@drm.service - Load Kernel Module drm was skipped because of an unmet condition check (ConditionKernelModuleLoaded=!drm)
.
[    8.177453] 10:13:08.129838 main     Executable: /usr/bin/VBoxDRMClient
[    8.186089] 10:13:08.139141 main     VBoxDRMClient: found compatible device: /dev/dri/renderD128
[    8.191180] 10:13:08.144040 main     VBoxDRMClient: starting deferred resize events delivery
[    8.194203] 10:13:08.147035 main     VBoxDRMClient: IPC server socket access granted to all users
[    8.443425] 10:13:08.396174 Timer    VBoxDRMClient: push screen layout data of 1 display(s) to DRM stack, fPartialLayout=false, rc=VERR_INV
[    8.444726] 10:13:08.397460 Timer    VBoxDRMClient: Error: Failed to push display change as requested by host, rc=VERR_INVALID_PARAMETER
[  301.978838] 10:18:02.078579 IpcCLT-1012 VBoxDRMClient: IPC client connection started
[  304.497497] 10:18:04.598027 IpcCLT-1012 VBoxDRMClient: thread IpcCLT-1012 notified IPC Client that display 0 is now primary, rc=VINF_SUCCES

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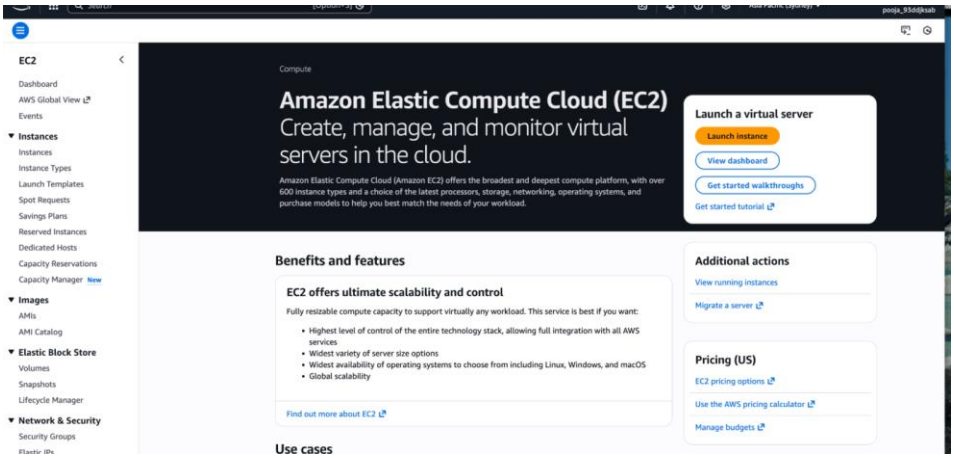

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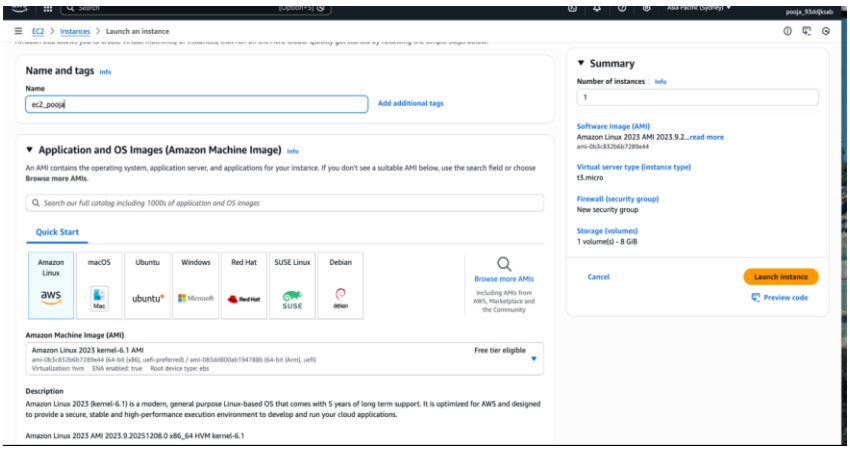

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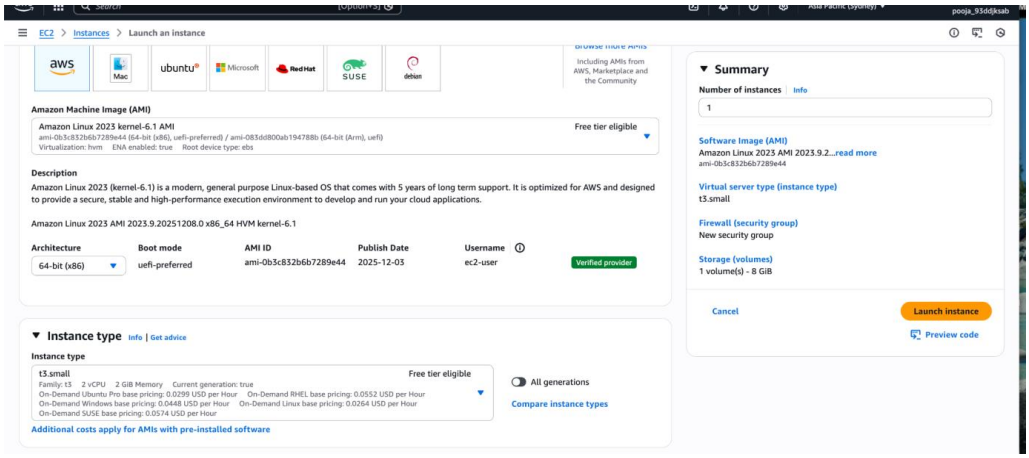

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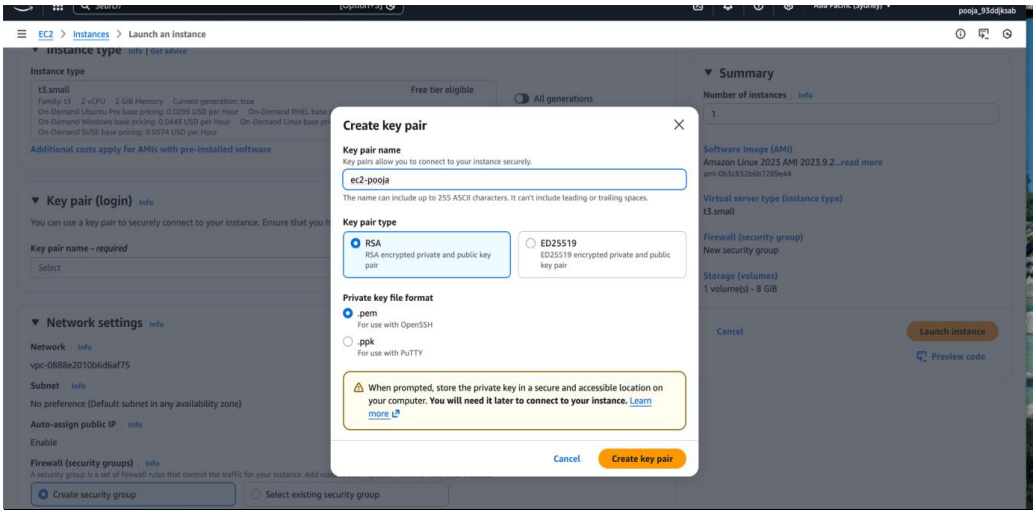

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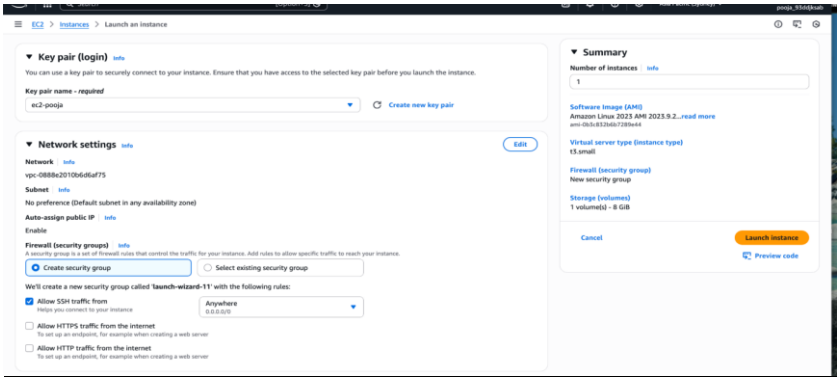

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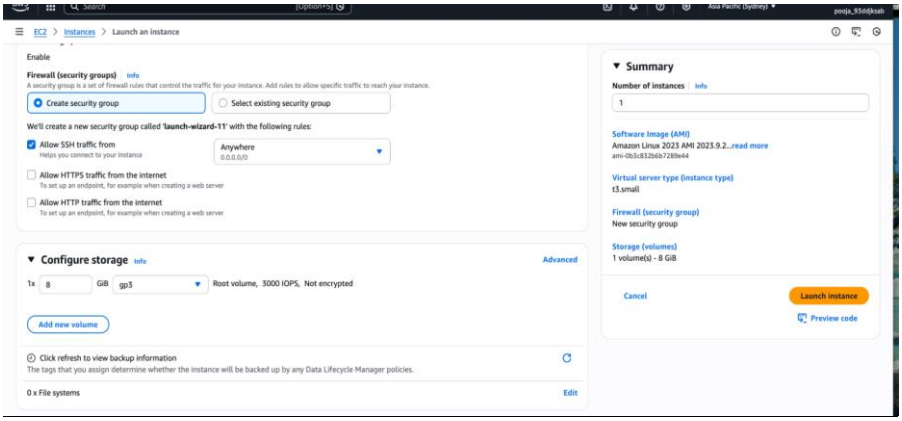

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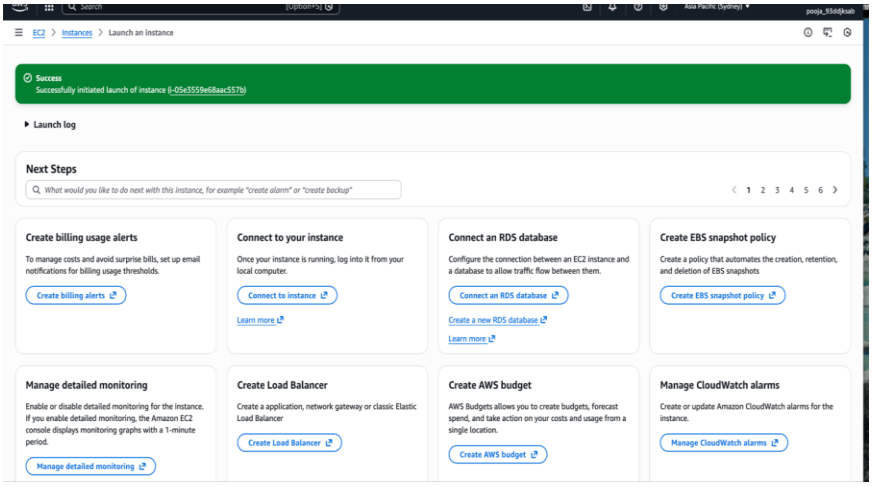

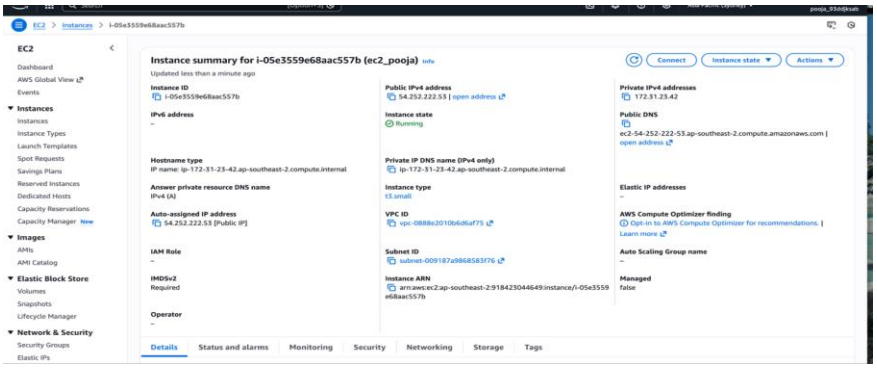

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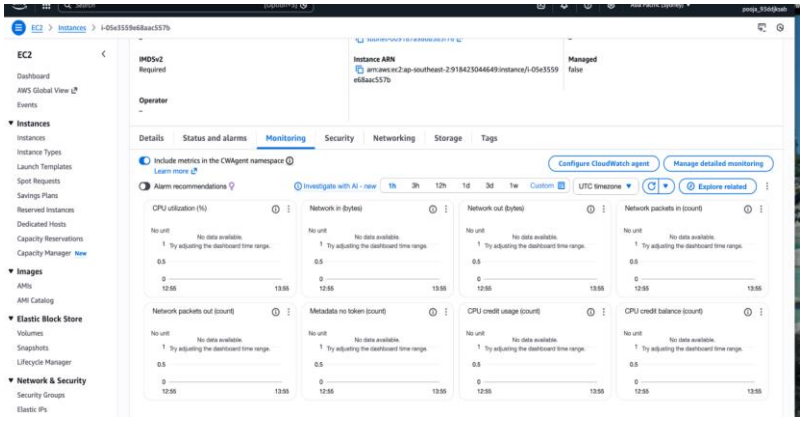

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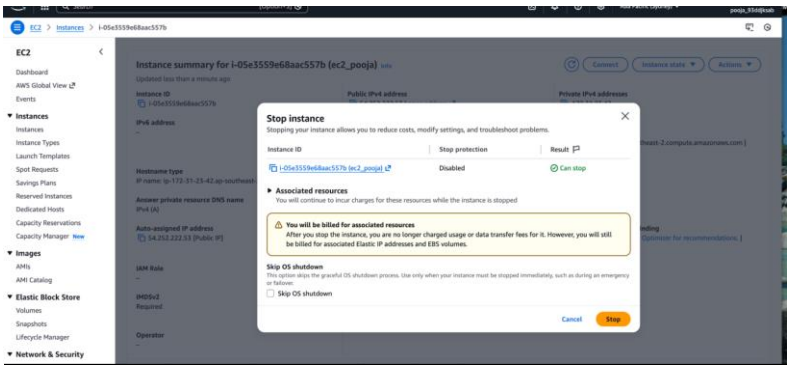

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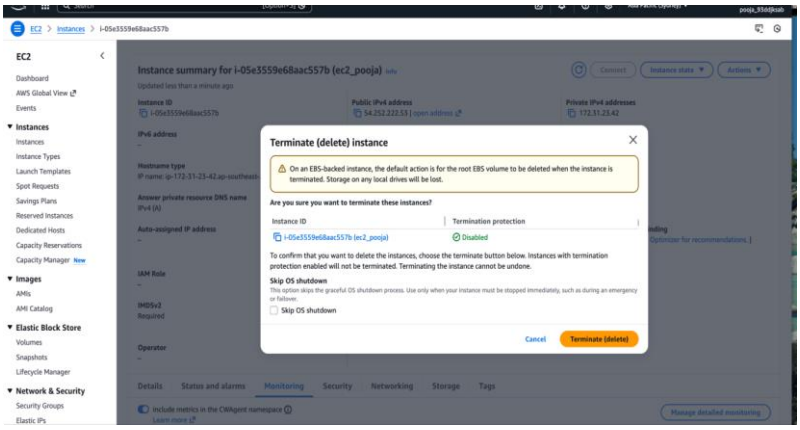

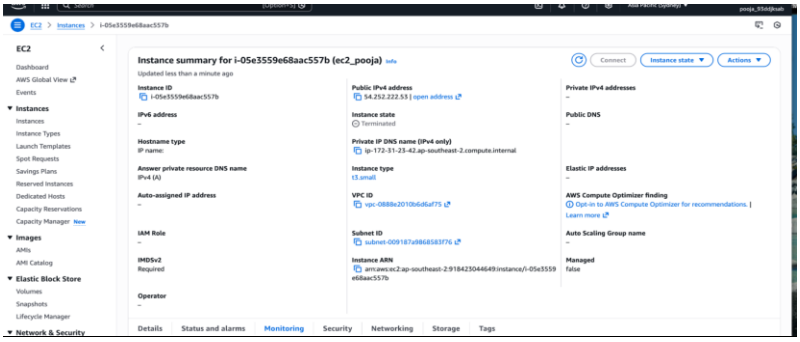

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